GTSC2143 Machine Learning for Business

Individual Assignment

Please finish this assignment via jupyter notebook and export it to a PDF or HTML file for submission. Clearly mark which part of the notebook fulfils which task. Please ensure that your jupyter notebooks do not have too much spurious output.

The use of generative AI tool is strictly prohibited for this assignment. If such use is detected, it will be considered as attempt at plagiarism.

Task 1 Regression on 'Reg' Dataset (50 points)

This exercise will test your ability to train, develop and deploy a Machine Learning regression model. The dataset consists of 27 features and a label (MaxTemp). Notice that several feature values are either missing or duplicated. A brief description of this data set can be found in file named Reg-Info.txt.

1. Perform Data Exploration:
a) Display the first few records in the dataset.
b) Display the number of rows and columns of the dataset.
c) Display the dataset statistics (min, max, ...)
d) Display the Null values of each feature.
e) Plot some graphs of the data to assist you in data exploration.

2. Perform initial cleaning of the data:
a) Delete columns which mainly contain Null values.
b) Remove duplicate columns (obvious redundant information).
c) Fill missing values in numeric columns if necessary.
d) Display the number of rows and columns of the data after cleaning.
e) Display the features left after cleaning.
f) Plot the distribution (histogram) of each of the following features: "MinTemp", "MeanTemp".

3. Analyze the pair-wise relationship between the features of the Data set.

4. Plot the correlation heatmap from the pairwise plots.

5. Perform necessary Data Preprocessing (Transformation) for final data preparation.
a) Scale values in numeric columns to a (0,1) range if needed.
b) Encode categorical data into one-hot vectors.
c) Split the dataset into training, validation and testing.

6. Choose the following Regression Models from Scikit Learn:
a) Simple Linear Regression.
b) Ridge Regression.
c) Lasso Regression.

7. Use only the features after cleaning in the dataset and obtain the following measures to compare the performance of the different models

- R2
- MAE
- MSE, RMSE

8. Perform Feature Selection (after cleaning the dataset), repeat the previous task (step 8), and again compare the different models.

Task 2 Classification on the 'credit-g' dataset (50 points)

Use the dataset with 'credit_g.csv'. The goal is to predict the "target" column.

1. Visualize the univariate distribution of each continuous feature, and the distribution of the target.
2. Split data into training and testing set.
3. Preprocess the data (such as treatment of categorical variables).
4. Choose appropriate classification algorithms and evaluate their performance with a training/validation split.
5. How different are the results among these models? Provide your explanation.
6. Point out the 5 most important features of the resulting models and explain why.